

Introduction to Mathematics and Optimization

June, 2021

PROBLEM 1

Let $f : U \rightarrow \mathbb{R}$ be the function given by

$$f(x, y) = x^3 + 2y^3,$$

where

$$U = \{(x, y) \in \mathbb{R}^2 \mid x < 1, y < 1, x + y > 1\}$$

- (a) Let $S = \mathbb{R}^2 \setminus U$. Write S using weak inequalities (i.e., inequalities with $\leq$ and $\geq$). Show that S is a union of closed sets.
- (b) Show that f is a strictly convex function. It may be used without proof that U is an open convex set.

Consider the optimization problem

$$\begin{aligned} \min f(x, y) \\ (x, y) \in C, \end{aligned} \tag{1}$$

where

$$C = \{(x, y) \in \mathbb{R}^2 \mid x \leq 1, y \leq 1, x + y \geq 1\}.$$

- (c) Explain, by a simple reference to the curriculum without calculations, why there must exist an optimal solution to (1).
- (d) Solve the optimization problem (1) using a method from the curriculum. Is the solution unique?

PROBLEM 2

Let

$$f(x, y, z) = x^2 + 2y^2 + z^2 - 2xy + 2yz$$

and

$$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

(a) Let

$$B = \begin{pmatrix} a & d & 0 \\ d & b & e \\ 0 & e & c \end{pmatrix},$$

where $a, b, c, d, e \in \mathbb{R}$. Show that

$$v^T B v = ax^2 + by^2 + cz^2 + 2dxy + 2eyz.$$

(b) Use (a) to find a symmetric 3×3 matrix A such that

$$\begin{aligned} f(x, y, z) &= x^2 + 2y^2 + z^2 - 2xy + 2yz \\ &= v^T A v. \end{aligned}$$

(c) Show that $f(x, y, z) \geq 0$ for all $x, y, z \in \mathbb{R}$.

(d) Find $x, y, z \in \mathbb{R}$ not all zero, such that $f(x, y, z) = 0$. Explain your method.

(e) Is f a strictly convex function?

(hint: Consider $u = 0 \in \mathbb{R}^3$ and $v \in \mathbb{R}^3$ with $v \neq 0$ and $f(v) = 0$. It may be used without proof that $f(tv) = t^2 f(v)$ for $t \in \mathbb{R}$)

PROBLEM 3

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y, z) = x^2 + y^2 + z^3 + xy + x + y - z.$$

- (a) Find the critical points of f .
- (b) Determine for each of the critical points whether they are local maxima, minima, or saddle points.
- (c) Is f a convex function?
- (d) Give an open convex set $U \subseteq \mathbb{R}^3$ such that f considered as a function on U is strictly convex.